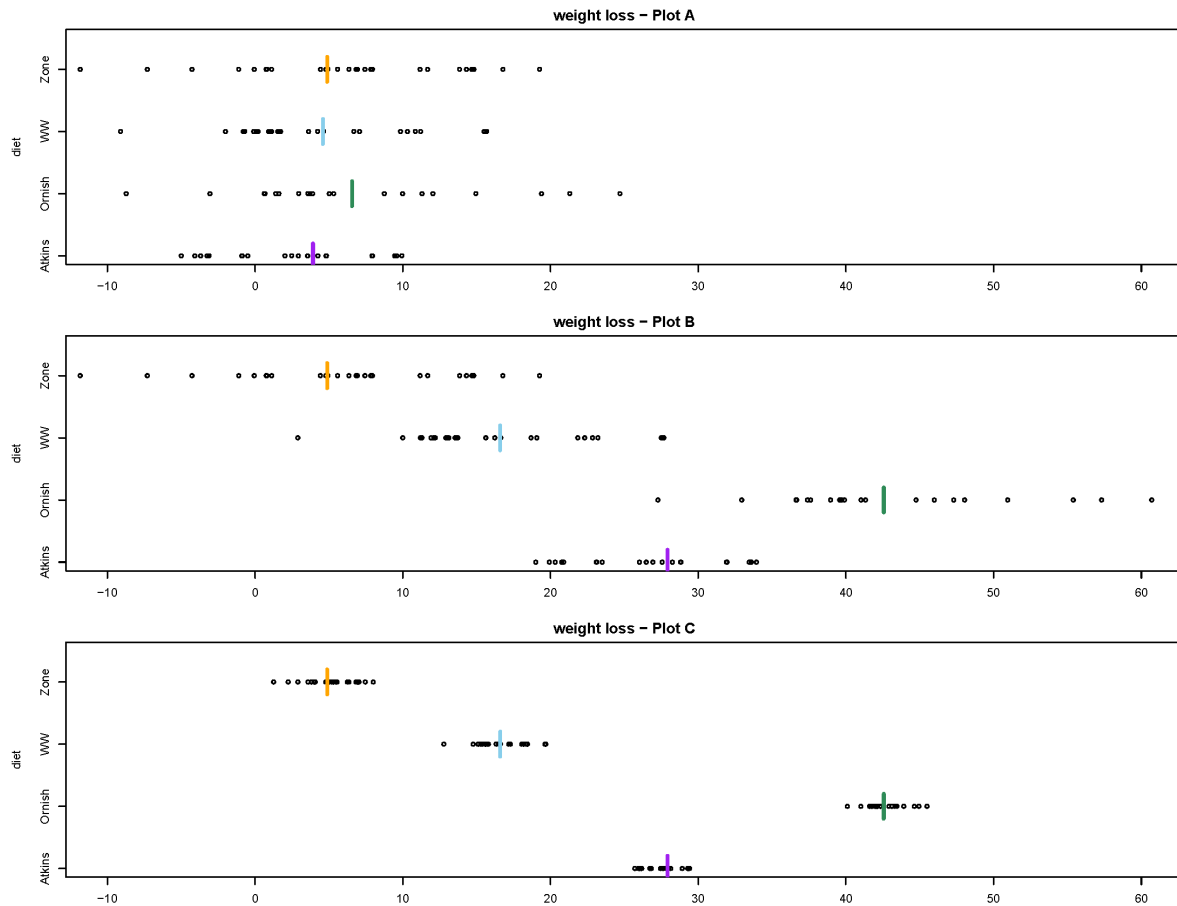


29 Comparing multiple means: ANOVA F procedures

When dealing with a numerical variable, we often summarize the variable with its mean. We have seen how to perform inference for a single mean and for comparing two means. Now we will see how to compare multiple means. The process is called “analysis of variance” or ANOVA, but that name is a little misleading. Our conclusions will still be about means.

Example 29.1. Consider the following plots, which are all on the same scale. (This example just illustrates some general ideas so the context isn’t important, but the data relate to a study on different types of diet plans (Atkins, etc) and weight loss. Plot A displays the study data. Plots B and C show hypothetical data sets with the same group sizes as in Plot A.) Group means are indicated by the colored vertical lines.

- The SD for each group in Plot B equals the SD for the corresponding group in Plot A.
- The mean for each group in Plot C equals the mean for the corresponding group in Plot B.



1. Rank the three scenarios (A, B, C) in order from weakest to strongest evidence to reject the null hypothesis of no difference in means between groups. That is, rank the plots in order from largest to smallest p-value.
2. Compare plots A and B.
 - The variability *between* groups in Plot B is (**choose one:** $>$, $<$, $=$) the variability *between* groups in Plot A.
 - The variability *within* groups in Plot B is (**choose one:** $>$, $<$, $=$) the variability *within* groups in Plot A.
3. Compare plots B and C.
 - The variability *between* groups in Plot C is (**choose one:** $>$, $<$, $=$) the variability *between* groups in Plot B.
 - The variability *within* groups in Plot C is (**choose one:** $>$, $<$, $=$) the variability *within* groups in Plot B.

4. Rank the plots in order from smallest to largest F statistic.

- Example 29.1 illustrates the idea behind the F statistic: we reject the null hypothesis of no difference in means between groups/populations/treatments if the observed variability *between* groups is large relative to the observed variability *within* groups.
- Roughly, we reject the null hypothesis if the ratio

$$\frac{\text{variability between groups}}{\text{variability within groups}}$$

is large enough. This idea is known as **analysis of variance (ANOVA)**.

- The name ANOVA refers to *what* we are doing, and not *why* we are doing it.
- Remember, we are analyzing the variance to make a conclusion about the *means*
- The F statistic quantifies the above ratio. The larger the value of the F statistic the stronger the evidence to reject the null hypothesis (of no difference) in favor of the alternative hypothesis (of some difference). That is, the larger the F statistic the smaller the p-value.

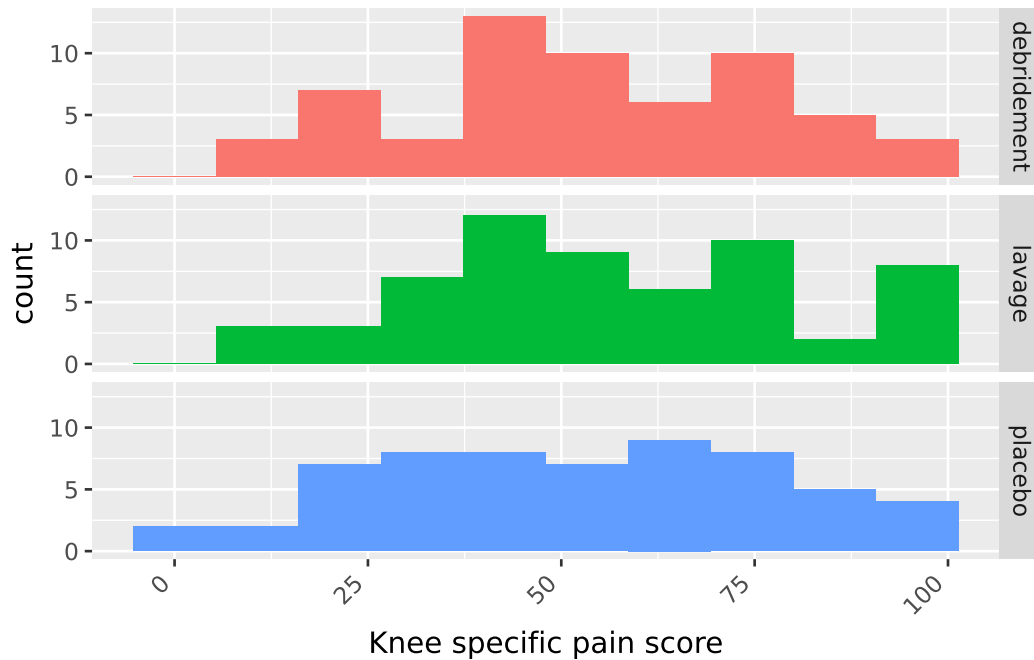
Example 29.2. When medical therapy fails to relieve the pain of osteoarthritis of the knee, arthroscopic lavage or debridement is often recommended. More than 650,000 such procedures are performed each year at a cost of roughly \$5,000 each. A recent double blind experiment investigated the effectiveness of these procedures¹. 180 subjects with osteoarthritis of the knee were randomly assigned to receive one of three treatments: arthroscopic lavage (which involved flushing the knee joint with fluid), arthroscopic debridement (which involved both lavage and shaving of cartilage and removal of debris), and placebo surgery (which involved incisions in the knee, but no insertion of instruments). Various outcomes were measured; the summaries below summarize scores two years after the procedure on a knee-specific pain scale (0-100, with higher scores indicating more severe pain). Does any procedure tend to result in better pain scores on average?

The following summarizes the sample data. Note that the sample size is the same within each group, which somewhat simplifies the computation of the ANOVA F statistic.

	count	mean	std	min	25%	50%	75%	max
treatment								
debridement	60.0	53.983333	23.304536	10.0	40.75	52.0	76.25	97.0

¹Source: <http://www.nejm.org/doi/full/10.1056/NEJMoa013259>

	count	mean	std	min	25%	50%	75%	max
treatment								
lavage	60.0	56.733333	24.150564	8.0	41.00	53.5	73.50	99.0
placebo	60.0	52.450000	25.079146	3.0	31.00	54.0	70.25	97.0



1. What statistic can be used to measure the variability *within* the Debridement group? Lavage? Placebo? How could you combine these numbers to to measure, in a single number, *variability within groups*?
2. If the null hypothesis is true, what would you expect about the group means?
3. What statistic can be used to measure the observed variability *between* groups?

4. Regardless of whether or not the null hypothesis is true, which of the values from parts 1 and 3 would you expect to be larger? Why?
5. How could we measure the expected variability between groups if the null hypothesis is true?
6. Compute the F statistic as the value from part 3 divided by the value from part 5.
7. What value would you expect F to be if the null hypothesis is true?
8. Specify in detail how you could use simulation to approximate the null distribution of the F statistic.
9. Figure 29.1 displays results from an applet used to run 10000 repetitions of the simulation. How could you use the simulation results to approximate the p-value? Could you use the empirical rule for Normal distributions?

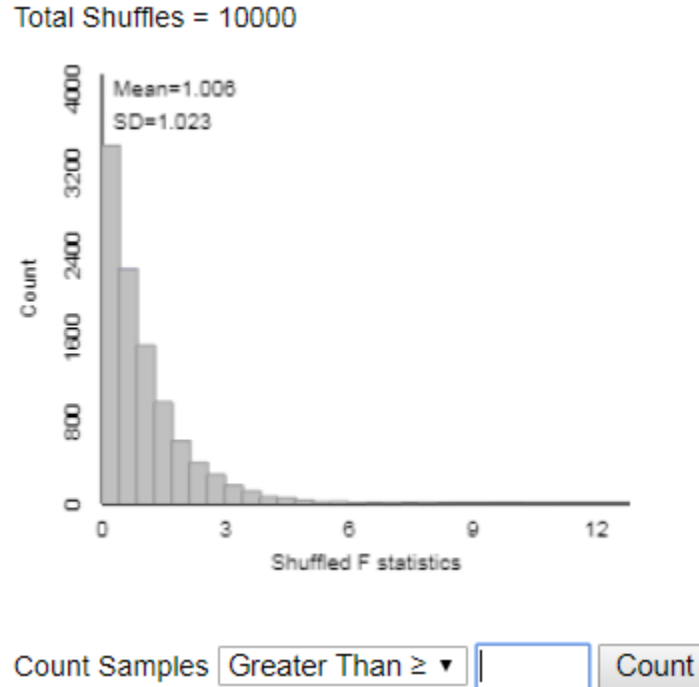


Figure 29.1: Permutation null distribution of F statistic for Example 29.2.

10. The p-value is about 0.6. State a conclusion in context. Are the results “significant”?

Hypothesis test for comparing multiple means: “One-way ANOVA F test”

- Situation: N observations of numerical response variable; categorical explanatory variable with g groups
- Parameters: μ_j 's are the population/process/treatment response means; σ is the common group SD of responses
- Goal: test for any differences in means
- The null hypothesis is $H_0 : \mu_1 = \dots = \mu_g$ (“no difference” or “no treatment effect”)
- The alternative hypothesis is not H_0 ; that is, there is some difference in means between groups, $H_1 : \mu_j \neq \mu_k$ for some $j \neq k$
- Test statistic: one-way ANOVA F statistic

$$F = \frac{\sum_{j=1}^g n_j (\bar{x}_j - \bar{x})^2 / (g - 1)}{\sum_{j=1}^g (n_j - 1) s_j^2 / (N - g)}$$

- The p-value is the probability of observing a value at least as large as the observed F based on an F distribution with $g - 1$ degrees of freedom in the numerator and $N - g$ degrees of freedom in the denominator
- Assumptions:
 - Either:
 - * The data are obtained through multiple unbiased and independent random samples from large populations
 - Or from one unbiased random sample which can be classified into multiple groups, and the groups can be considered independent of each other.
 - * The data are obtained from an experiment in which subjects were to two treatments.
 - No bias in data collection
 - Comparing *means* is an appropriate analysis
 - Equal variance: True SD of responses is same, σ , for each population/process/treatment.
 - * Typically satisfied through random assignment.
 - * Might not be satisfied if the data are sampled from different populations/processes.
 - * One rule of thumb is that the largest observed sample SD should be no more than twice the smallest.
- Special case: when the group size n is the same for all g groups is

$$F = \frac{\text{group size} \times \text{variance of group means}}{\text{mean of group variances}}$$

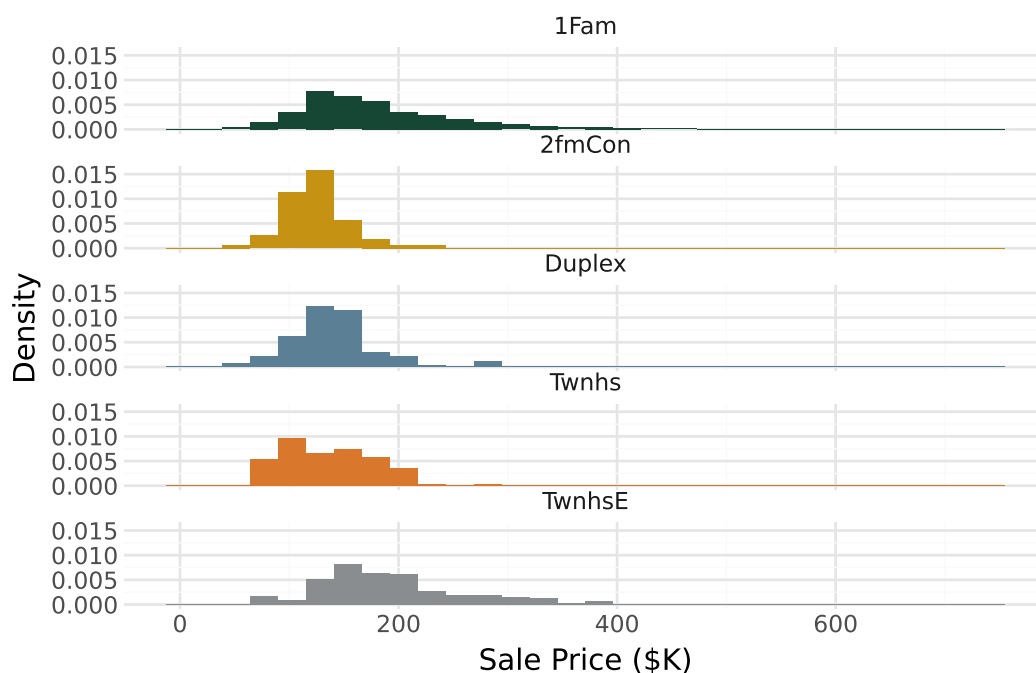
$$= \frac{n \sum_{j=1}^g (\bar{x}_j - \bar{x})^2 / (g - 1)}{\sum_{j=1}^g s_j^2 / g}$$

- The calculations are traditionally summarized in an **ANOVA table**

Source	df	SS (Sum of squares)	MS (Mean Square)	F
Between groups (a.k.a. “treatment”)	$g - 1$	$SS(\text{between}) = \sum_{j=1}^g n_j (\bar{x}_j - \bar{x})^2$	$\frac{SS(\text{between})}{g - 1}$	$\frac{MS(\text{between})}{MS(\text{within})}$
Within groups (a.k.a. “error”)	$N - g$	$SS(\text{within}) = \sum_{j=1}^g (n_j - 1) s_j^2$	$\frac{SS(\text{within})}{N - g}$	
Total	$N - 1$	$(SS(\text{between}) + SS(\text{within})) = \sum_{i=1}^N (x_i - \bar{x})^2$		

Example 29.3. Continuing with the Ames housing data set. Now suppose we want to compare sale price of homes by building type: single-family (1Fam), two-family (2fmCon), duplex, townhouse end unit (TwnhsE), townhouse inside unit (Twnhs). The following summarizes the sample data.

BldgType	count	mean	std	min	25%	50%	75%	max
1Fam	2425.0	184.812041	82.821802	12.789	130.0000	165.000	220.000	755.00
2fmCon	62.0	125.581710	31.089240	55.000	106.5625	122.250	140.000	228.95
Duplex	109.0	139.808936	39.498974	61.500	118.8580	136.905	153.337	269.50
Twnhs	101.0	135.934059	41.938931	73.000	100.5000	130.000	170.000	280.75
TwnhsE	233.0	192.311914	66.191738	71.000	145.0000	180.000	222.000	392.50



1. Table 29.1 is the ANOVA table. The F statistic is 26.1 and the p-value is basically 0. State the conclusion of the ANOVA F test in context.
2. What are some natural follow-up questions? How might you use the data to answer them?

Table 29.1: ANOVA table for Example 29.4.

	sum_sq	df	F	PR(>F)
C(BldgType)	6.454111e+05	4.0	26.151358	2.475923e-21
Residual	1.804713e+07	2925.0	NaN	NaN

- Rejecting the null hypothesis based on an F statistic only tells you that “the mean for at least one of the treatments (populations) is not the same as the mean for at least one of the other treatments (populations)”.
- The natural follow-up questions are: which treatment (population) means differ? and by how much?
- One approach when you have rejected the null hypothesis is to calculate — using software — all pairwise confidence intervals. In particular, which intervals do not contain zero?
- When computing multiple confidence intervals, the multiple (z^* or t^*) should be increased² to account for the **issue of multiple comparisons** (recall Exercise 26.2 and related notes).
- **Tukey’s method** is a widely used method for constructing simultaneous confidence intervals for *all pairwise comparisons of means*. The confidence intervals computed using Tukey’s method already contain an adjustment to account for multiple comparisons.

Example 29.4. Continuing Example 29.3. Table 29.2 contains the results of Tukey pairwise 95% confidence intervals.

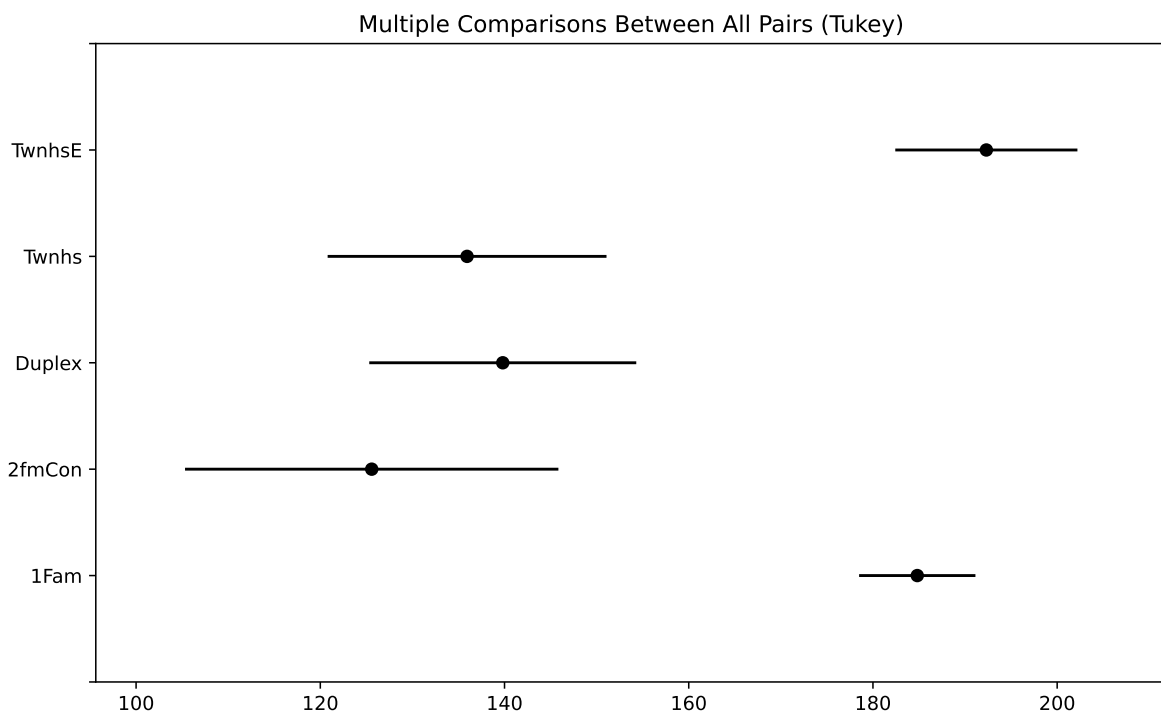
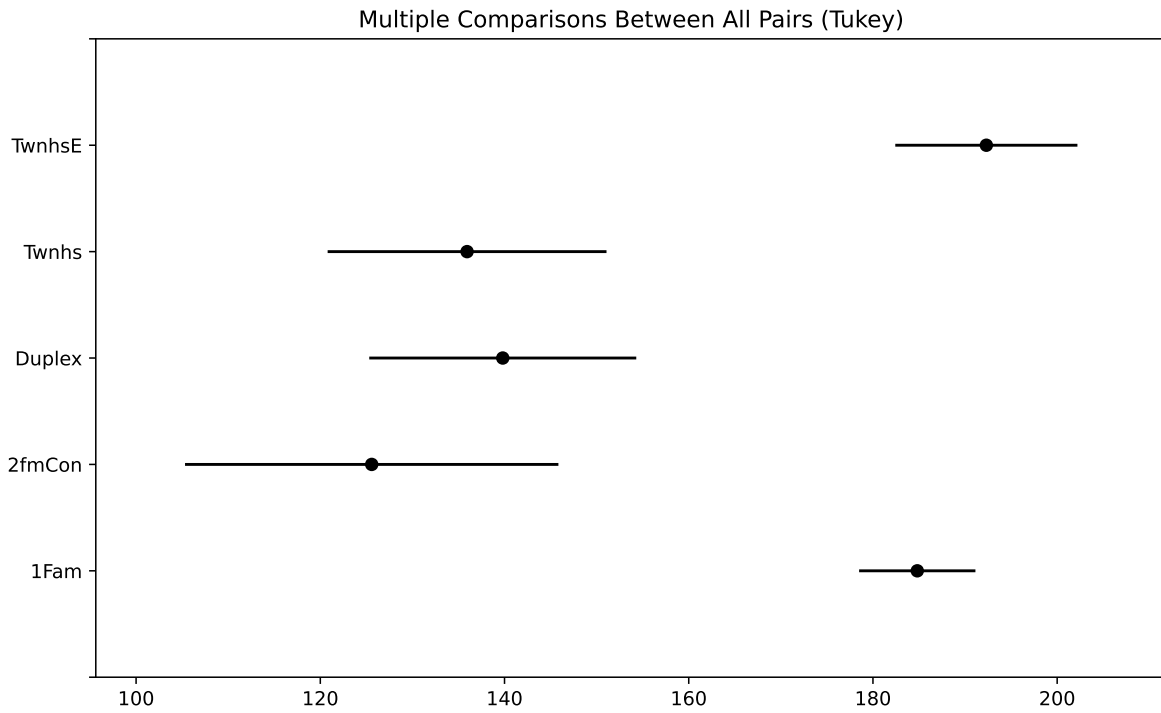
1. For which pairs of building types is there reasonable evidence of a difference in mean sale price? For which pairs is there NOT reasonable evidence of a difference?

²There are several ways to do this. The simplest is a *Bonferroni* adjustment, which splits the error rate evenly across all intervals. For example, for simultaneous 95% confidence in 10 intervals (i.e. a total error rate of 0.05), use the t^* factor for 99.5% confidence ($0.995 = 1 - 0.05/10$), $t^* = 2.8$, instead of 2 when computing the intervals. But the Tukey method is more widely used when comparing means.

Table 29.2: Tukey pairwise 95% confidence intervals for Example 29.4.

Multiple Comparison of Means - Tukey HSD, FWER=0.05						
group1	group2	meandiff	p-adj	lower	upper	reject
1Fam	2fmCon	-59.2303	0.0	-86.8048	-31.6559	True
1Fam	Duplex	-45.0031	0.0	-65.9952	-24.0111	True
1Fam	Twnhs	-48.878	0.0	-70.6511	-27.1049	True
1Fam	TwnhsE	7.4999	0.6328	-7.2051	22.2048	False
2fmCon	Duplex	14.2272	0.786	-19.8771	48.3316	False
2fmCon	Twnhs	10.3523	0.9255	-24.2382	44.9429	False
2fmCon	TwnhsE	66.7302	0.0	36.0924	97.368	True
Duplex	Twnhs	-3.8749	0.9965	-33.4861	25.7363	False
Duplex	TwnhsE	52.503	0.0	27.6234	77.3825	True
Twnhs	TwnhsE	56.3779	0.0	30.8358	81.9199	True

- Interpret in context each of the confidence intervals that does NOT contain 0.



29.1 Exercises

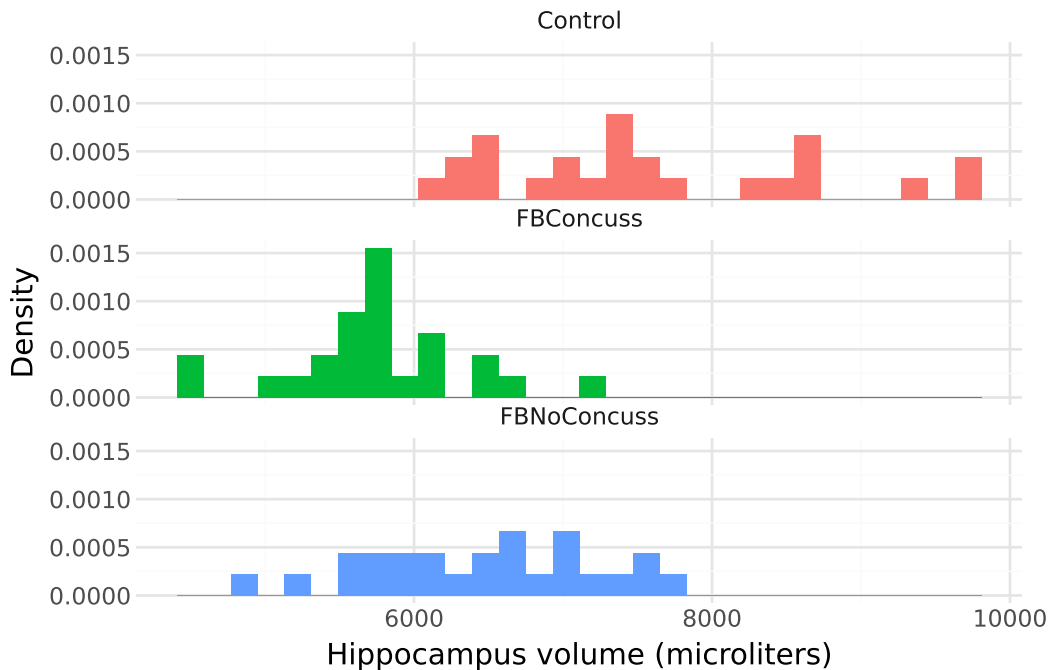
Exercise 29.1. Continuing Example 29.2. If we were to compute a series of pairwise confidence intervals for the difference in group means — Debridement — Lavage, Debridement — Placebo, Lavage — Placebo — what value would each of the confidence intervals contain? Why?

Exercise 29.2. A study³ on brain trauma in football players included 3 groups, with 25 cases in each group. The control group consisted of healthy individuals with no history of brain trauma who were comparable to the other groups in age, sex, and education. The second group consisted of NCAA Division 1 college football players with no history of concussion, while the third group consisted of NCAA Division 1 college football players with a history of concussion. High resolution MRI was used to collect brain hippocampus volume (microliters).

Plots and summary statistics of hippocampus volume (microliters) for all three groups

	count	mean	std	min	25%	50%	75%	max
Group								
Control	25.0	7602.6	1074.022928	6175.0	6780.0	7385.0	8510.0	9710.0
FBConcuss	25.0	5734.6	593.397562	4490.0	5505.0	5710.0	6035.0	7160.0
FBNoConcuss	25.0	6459.2	779.739112	4810.0	5965.0	6515.0	7020.0	7790.0

³Source: Singh R, Meier T, Kuplicki R, Savitz J, et al., “Relationship of Collegiate Football Experience and Concussion With Hippocampal Volume and Cognitive Outcome,” JAMA, 311(18), 2014.



1. Specify in words the null hypothesis.
2. The SD of the numbers 7602.6, 6459.2, and 5734.6 is 941.8. Compute by hand the ANOVA F statistic.
3. The ANOVA F statistic is 31.5. Describe in full detail how (in principle) you could use index cards to simulate the null distribution of the ANOVA F statistic and how you would use the simulation results to find the p-value. (You don't have to compute the p-value; just explain how you could find it after you performed the simulation.)
4. Use [this applet](#) to conduct the simulation from the previous part.
 - You will first need to copy and paste [the data](#) from this study into the applet.
 - In the Select Statistic box in the lower left, select F-statistic. Check that the Observed F-statistic is 31.473, and that your summary statistics match those from the summaries.
 - Click the "Show Shuffle Options" box and then run the simulation.
 - Enter the appropriate value in the "Count Samples" box and click "Count" to compute the p-value.
5. The p-value is basically 0 (see Table 29.3). Write a clearly worded sentence containing the conclusion of the hypothesis test in the context of the problem.
6. Table 29.4 contains the Tukey confidence intervals. Which of the confidence intervals contain 0, and which do not? Explain what this means.
7. Write a clearly worded sentence interpreting each of the three confidence intervals in context.

Table 29.3: ANOVA table for Exercise [29.2](#).

	sum_sq	df	F	PR(>F)
C(Group)	44348606.0	2.0	31.473165	1.507201e-10
Residual	50727336.0	72.0	NaN	NaN

Table 29.4: Tukey pairwise 95% confidence intervals for Exercise [29.2](#).

Multiple Comparison of Means - Tukey HSD, FWER=0.05						
group1	group2	meandiff	p-adj	lower	upper	reject
Control	FBConcuss	-1868.0	0.0	-2436.1525	-1299.8475	True
Control	FBNoConcuss	-1143.4	0.0	-1711.5525	-575.2475	True
FBConcuss	FBNoConcuss	724.6	0.0088	156.4475	1292.7525	True